

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause personal injury.

⚠ WARNING ⚠

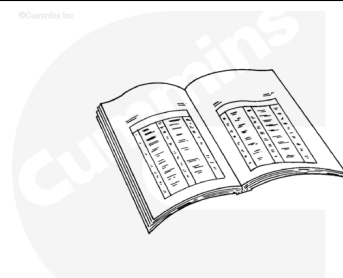
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



ck800wa

Note : The illustrations shown will be the SS model single cylinder air compressor. Differences in procedures for SS, QE, and ST Cummins® air compressors will be shown where necessary.

- Use steam to clean the air compressor. Dry with compressed air.
- Remove the fuel pump. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html>)
- Drain the engine coolant. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008->

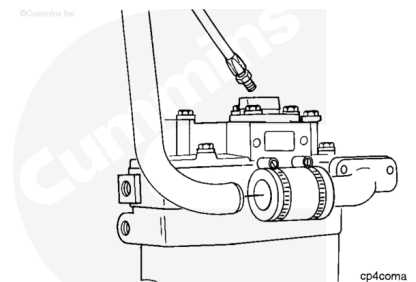
018-tr.html)

- Remove the coolant lines from the air compressor.

Remove

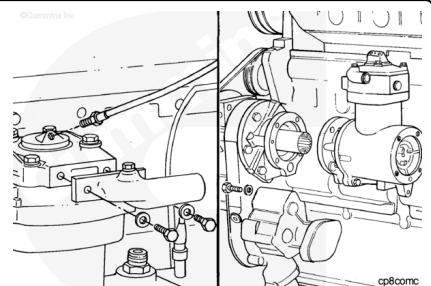
Single Cylinder

Remove the air inlet and outlet connections from the air compressor.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the air connections from the air compressor.

Remove the air compressor support bracket and capscrews.

Remove the four capscrews, the air compressor, and splined coupling.

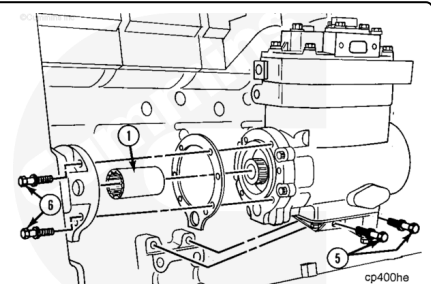
Twin Cylinder

Remove the air compressor support mounting capscrews (5).

Remove the four capscrews (6). Remove the air compressor.

Remove the splined coupling (1). Remove and discard the gasket.

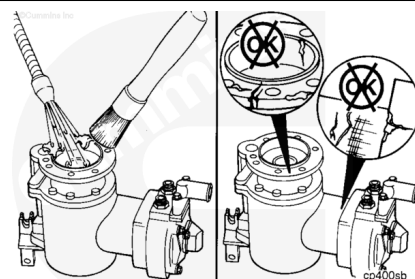
If a two-piece bracket is used, remove the bolts and nuts securing the brackets together. Remove the bracket from the compressor and the bracket from the block.



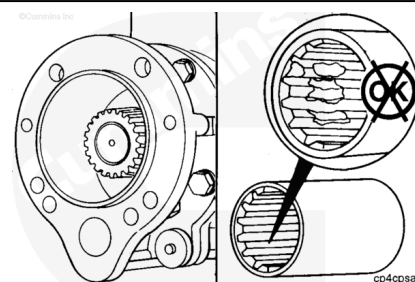
Clean and Inspect for Reuse

Single Cylinder

Clean air compressor and inspect housing for cracks or damage.



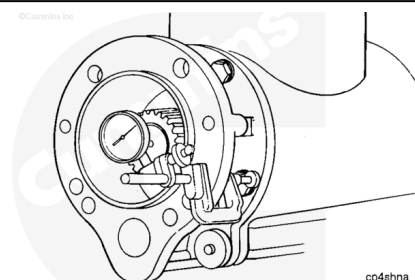
Inspect the splined coupling for cracks or damage.



Check crankshaft end clearance on the single cylinder air compressor.

End Clearance (Aluminum Support, Part Number 3005153)

mm		in
.05	MIN	.002
.69	MAX	.027

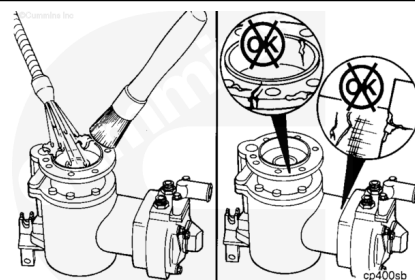


If end clearance is **not** within specifications, go to the Master Repair Manual, Holset® Air Compressors, Bulletin 3666121.

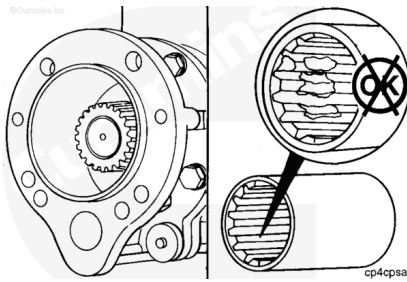
Twin Cylinder

Use solvent, Part Number 3824421, or equivalent, and clean the exterior of the air compressor.

Inspect the compressor housing for cracks or damage.



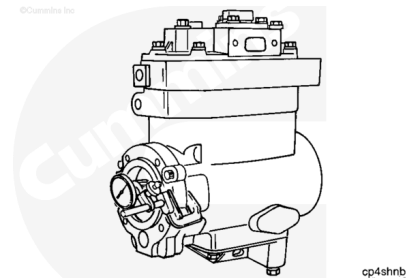
Inspect the splined coupling for cracks or damage.



Check the crankshaft end clearance on the two cylinder air compressor.

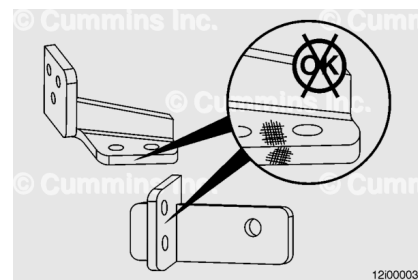
Crankshaft End Clearance		
mm		in
0.038	MIN	0.0015
0.200	MAX	0.0080

If end clearance is **not** within specifications, go to ST676 Two Cylinder Air Compressor Shop Manual, Bulletin 3666221, for rebuild instructions.



Inspect the brackets and capscrews for fretting or other damage. If damage is found, the brackets **must** be replaced to prevent damage to the air compressor housing.

Note : Surface wear that **only** removes the paint on the brackets is acceptable.

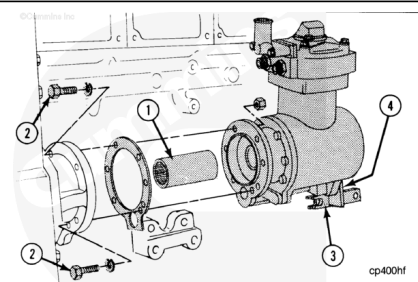


Install

Single Cylinder

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



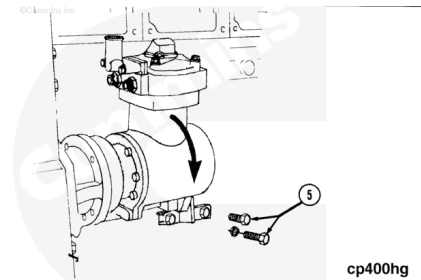
Install the splined coupling (1) on the accessory drive shaft.

Be sure the gasket surfaces of the accessory drive and air compressor are clean and **not** damaged.

Use a new gasket to install the air compressor.

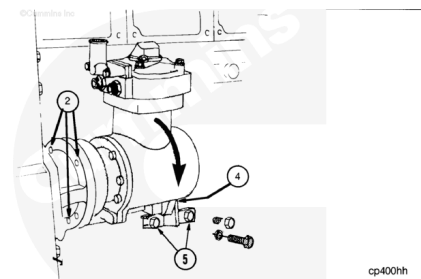
Install and **only** hand-tighten the four capscrews and nuts (2).

Install the two capscrews (5) for the bracket. The support bracket **must** be flat against the block. Turn the compressor until the bracket is flat against the block.



⚠ CAUTION ⚠

The bracket must be flat against the compressor. Failure to have the bracket flat can cause premature component failure.



Tighten the compressor to the accessory drive capscrews (2).

Torque Value: 60 n•m [44 ft-lb]

Tighten the bracket to the block capscrews (5).

Torque Value: 45 n•m [33 ft-lb]

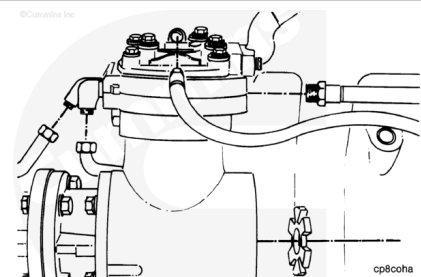
Tighten the bracket to the compressor capscrew (4).

Torque Value: 45 n•m [33 ft-lb]

Note : If rubber grommets are used on the coolant lines, be sure they are installed carefully to prevent cuts or tears to the grommets which cause coolant leaks. When flexible tubing is used, make sure that it does **not** rub any other surface.

Install the coolant and air lines to the air compressor and tighten.

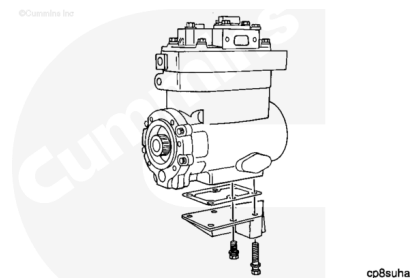
Install the fuel pump drive coupling on the air compressor.



Twin Cylinder

⚠ WARNING ⚠

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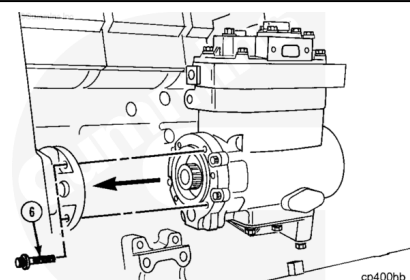
On twin cylinder compressors with a one piece bracket, install the gasket and the cover plate (support bracket) on the compressor.

Do **not** tighten the six capscrews. The support **must** be adjusted.

Note : Do not install the splined coupling or the mounting gasket.

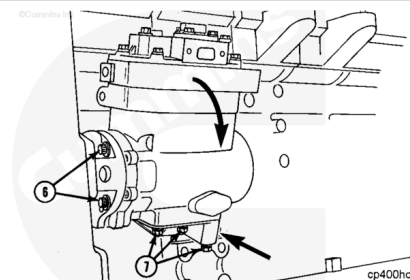
Install the compressor on the accessory drive.

Install the four capscrews (6). **Only** tighten the capscrews enough to pull the compressor to the accessory drive. The compressor **must** be turned to align the support bracket.



⚠ CAUTION ⚠

The support must be flat against the block. Rotate the compressor until the support is aligned properly against the block. If the support is not flat, the compressor and/or accessory drive will fail.



Tighten the four capscrews (6).

Torque Value: 60 n•m [44 ft-lb]

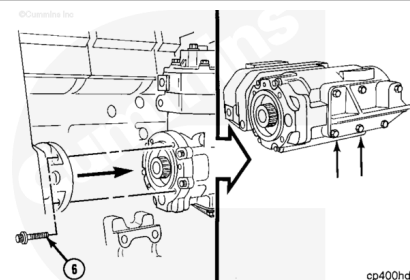
The bracket **must** remain flat against the block. Tighten the support capscrews (7) that are accessible.

Torque Value: 45 n•m [33 ft-lb]

Remove the four capscrews (6). Remove the compressor.

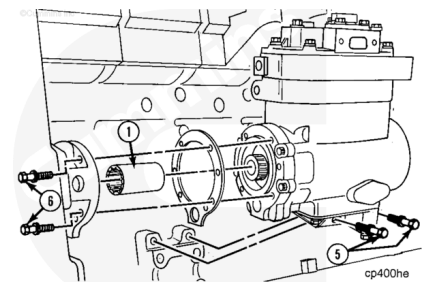
Tighten the two remaining supports to the compressor capscrews.

Torque Value: 45 n•m [33 ft-lb]



⚠ CAUTION ⚠

The support must be flat against the block. Rotate the compressor until the support is aligned properly against the block. If the support is not flat, the compressor and/or accessory drive will fail.



Install the splined coupling (1), gasket, and compressor.

Install the four capscrews (6) and two capscrews (5).

Torque Value:

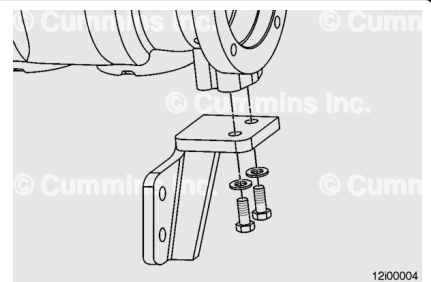
Capscrew (6) 60 n•m [44 ft-lb]

Torque Value:

Capscrew (5) 45 n•m [33 ft-lb]

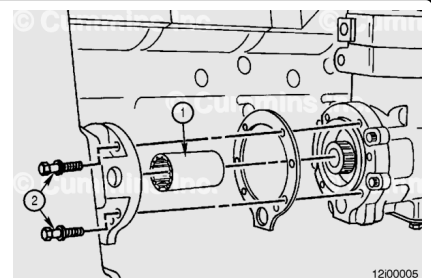
On air compressors with the two-piece bracket, install the top bracket, washers, and capscrews onto the bottom face of the air compressor.

Hand tighten the capscrews.



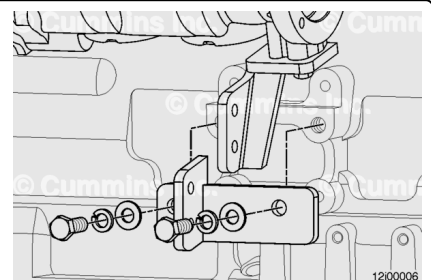
Install the splined coupling (1), gasket, and compressor.

Install the four capscrews (2), but do **not** tighten to allow movement of the compressor for support bracket installation.



Position the bottom bracket behind the top bracket and install the capscrews, lock washers, and plain washers.

Hand-tighten the capscrews.

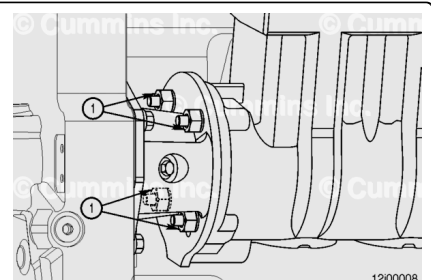


Tighten the four air compressor mounting capscrews (1).

Torque Value:

Mounting Capscrews 47 n•m [35 ft-lb]

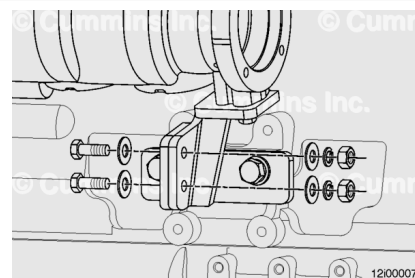
Note : The air compressor mounting brackets have slotted holes to allow for proper alignment when the air compressor is tightened.



Install the two capscrews through both brackets using two plain washers, one lock washer, and one nut on each capscrew.

Hand-tighten the capscrews and nuts.

Note : Be sure the bracket mating surfaces are flush before tightening capscrews.



Note : Be sure brackets are flush against the compressor, block, and each other.

Tighten the capscrews to the compressor (1).

Torque Value:

Compressor Capscrews 47 n•m [35 ft-lb]

Tighten the capscrews to the block (2).

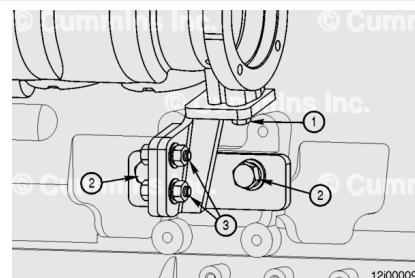
Torque Value:

Capscrews to Block 135 n•m [100 ft-lb]

Tighten the capscrews and nuts (3).

Torque Value:

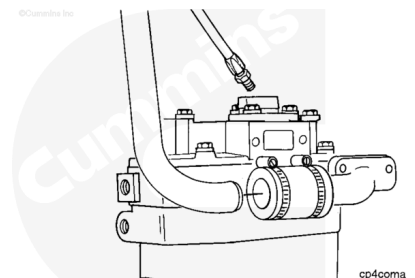
Capscrews and Nuts 47 n•m [35 ft-lb]



On the one-piece and two-piece brackets, install the air inlet and outlet connections to the air compressor.

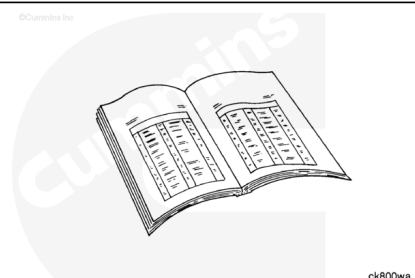
Tighten the clamps.

Torque Value: 6 n•m [53 in-lb]



Finishing Steps

- Install the fuel pump. Refer to Procedure 005-016 in Section 5. (</qs3/pubsys2/xml/en/procedures/20/20-005-016-tr.html>)
- Install the coolant lines to the air compressor.
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/20/20-008-018-tr.html>)



- Operate the engine to operating temperature and check for leaks.

Last Modified: 11-Nov-2010
